



USB-C ENTRY

Pin configuration diagram for the BQ24073RGTR IC. The IC is shown as a central rectangle with pins numbered 1 through 16. Pin 1 is BAT_NTC, Pin 2 is VBAT, Pin 3 is BAT, Pin 4 is CE, Pin 5 is EN2, Pin 6 is EN1, Pin 7 is USB_PRESENT_N, Pin 8 is CHG_N, Pin 9 is CHG, Pin 10 is VSYS, Pin 11 is OUT, Pin 12 is ILIM, Pin 13 is VBUS, Pin 14 is TMR, Pin 15 is CHG_TD, and Pin 16 is CHG_ISET. The IC is labeled BQ24073RGTR at the bottom right.

Diagram illustrating the connection for the J102 connector. The J102 connector has three pins: 1 (BAT_NTC), 2 (BAT_RAW), and 3 (BAT_RAW). The BAT_RAW signal is connected to the R108 resistor, which is then connected to the BAT_IN pin.

Pin configuration diagram for the MAX17048G+T10 (U103). The chip is shown with its pins and connections:

- VDD** (Pin 3) connected to **VBAT**.
- CELL** (Pin 2) connected to **VBAT**.
- CTG** (Pin 1) connected to **VBAT**.
- QSTRT** (Pin 6) connected to **VBAT**.
- SDA** (Pin 8) connected to **I2C_SDA**.
- SCL** (Pin 7) connected to **I2C_SCL**.
- ALRT** (Pin 5) connected to **SENSE_ALARM_N**.
- GND** (Pin 4) connected to ground.
- EP** (Pin 9) connected to ground.

Sheet: /Power/

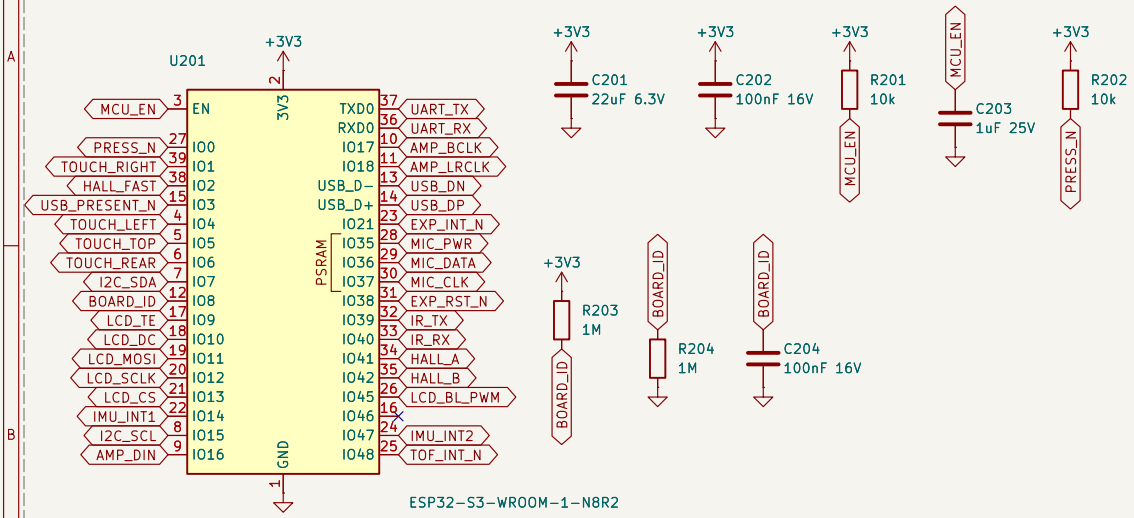
Title: MAO MAIN A0 - Power

Size: A3	Date: 2020-10-04	PWR_FLAG	REV: A0
KiCad EDA 10.0.6	GND	VBUS	Id: 3/6

RISC0 E.D.A. 10.0.0							10. 2/0
			7				8

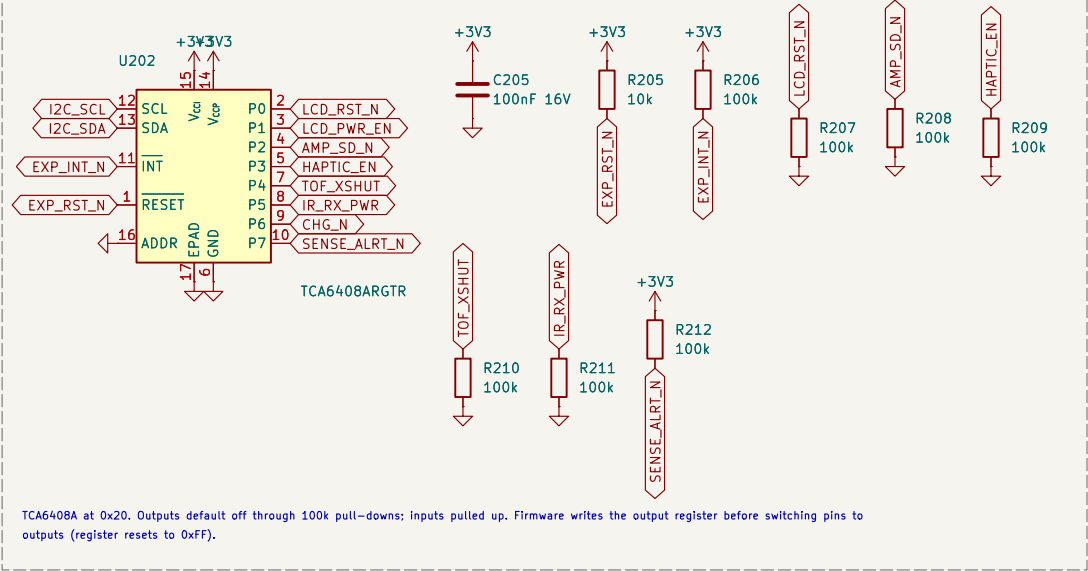
COMPUTE

MCU



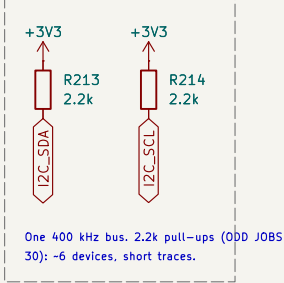
ESP32-S3-WROOM-1-NBR2 on B,Cu at 6 o'clock, antenna over the board-edge notch (no copper beneath). 22 uF + 100 nF at 3V3. EN: 10k/1 uF RC. GPIO0 = face press (B00T). Unused pins NC or test pads.

EXPANDER



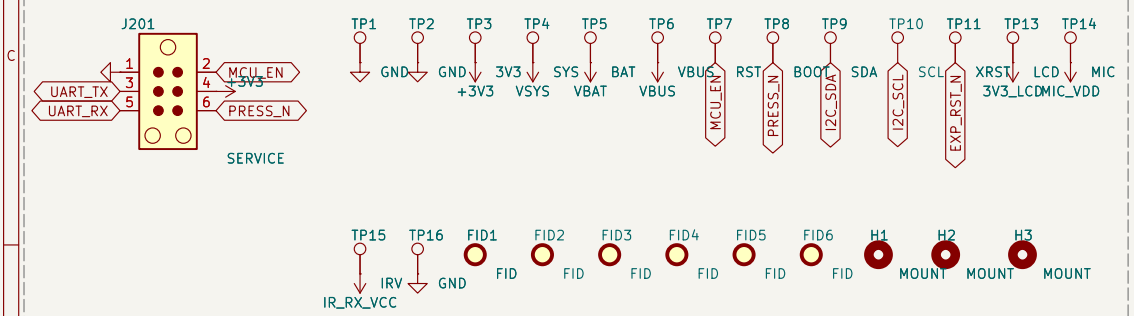
TCA6408A at 0x20. Outputs default off through 100k pull-downs; inputs pulled up. Firmware writes the output register before switching pins to outputs (register resets to 0xFF).

I2C



One 400 kHz bus. 2.2k pull-ups (ODD JOBS 30): -6 devices, short traces.

SERVICE



Tag-Connect TC2030-NL (no part fitted): 1 GND, 2 EN, 3 TXD0, 4 3V3, 5 RXD0, 6 GPIO0 - recovery without USB (ODD JOBS 33-35, 157). Pinout chosen so TXD0/RXD0 run straight from the module pins. Probe pads: rails beside the charger, comms in one field, switched rails at their sources (36-39).

ESP32-S3 module, GPIO expander, boot/reset, service and test access

ODD JOBS

Sheet: /Compute/

File: compute.kicad_sch

Title: MAO_MAIN A0 - Compute

Size: A3

Date: 2026-10-04

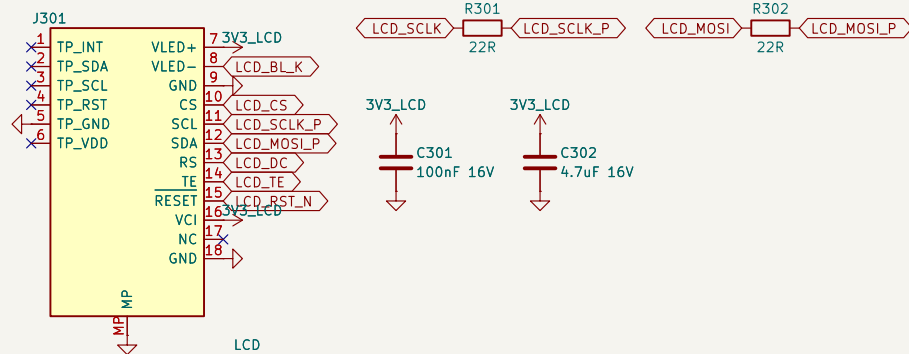
Rev: A0

KiCad E.D.A. 10.0.6

Id: 3/6

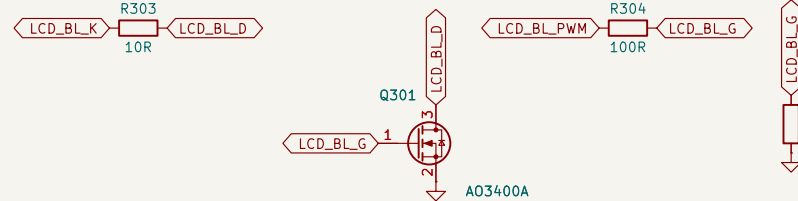
INTERFACE

DISPLAY



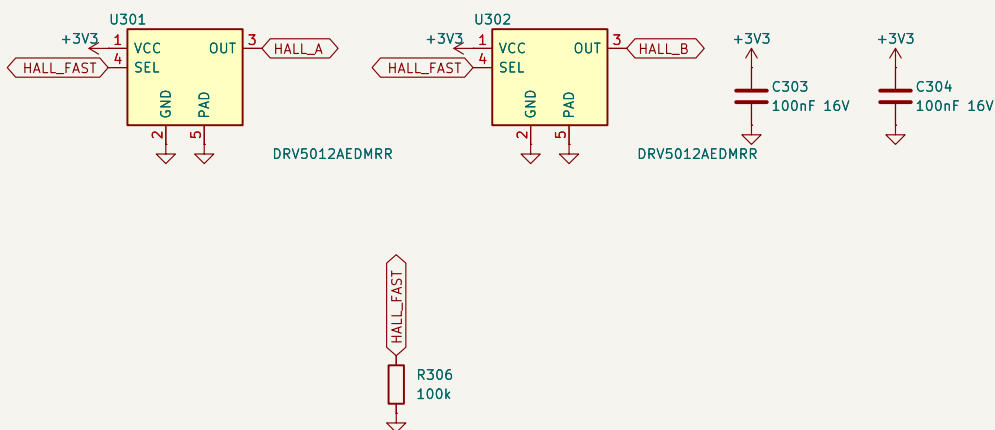
Plug-in display: 1.28" round GC9A01 panel (Winstar WF0128BTAAADNNO, 240x240 IPS) on an 18-pin 0.5 mm FPC tail, folded under the panel into a 1.0 mm high back-flip connector with top and bottom contacts, so any panel of this 18-pin standard plugs in either way and can be swapped without solder. Pins 1-6 serve the touch controller of touch variants: not wired (MAO's face is a pressed window), TP_GND to GND, TE to GPIO9 for tear-free frames, 22R on SCLK/MOSI damps 80 MHz edges (ODDS JVB 29). Panel and backlight on the switched 3VS1_LCD rail; RST held low until firmware.

BACKLIGHT



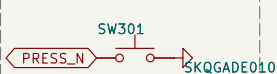
Two parallel white LEDs, VLED 2.8–3.2 V (3.0 typ) at 40 mA (panel spec). Low-side AO3400A, PWM on GPIO45. 10R from 3V3_LCD sets ~30 mA. Gate pull-downs: dark until firmware (ODD JOBS 116).

RING DIAL



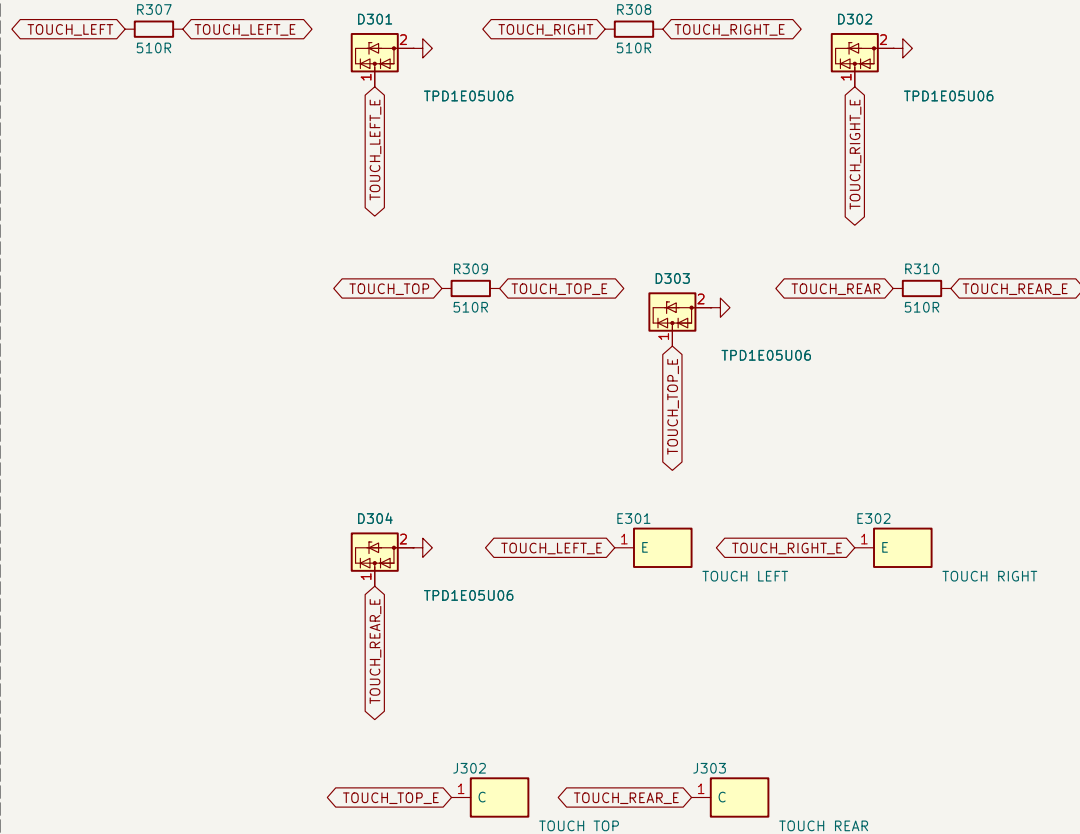
Two DRV5012 Hall latches under the ring's 30-pole ferrite strip, 6 deg apart (half a pole = 90 deg electrical): A/B quadrature identical to the LCDkit EC11 (2 transitions per detent, 30 detents). HALL_FAST low = 20 Hz sampling, 1.6 uA (sleep); high = 2.5 kHz (awake).

FACE PRESS



ALPS SKQG on F.Cu at the centre, under the panel: the floating face presses it.
GPIO0 = BOOT strap: no capacitor here.

BODY TOUCH



Four ESP32-S3 touch channels. 510R within 1 mm of the module (Espressif guide); 0.42 pF ESD at the electrode. LEFT/RIGHT: copper arcs at the rim. TOP/REAR: gold springs to printed or foil electrodes.

Round display, backlight, ring dial, face press, body touch

ODD JOBS

Sheet: /Interface/

File: interface.kicad_sch

Title: MAO_MAIN A0 – Interface

Size: A3

Date: 2026-10-04

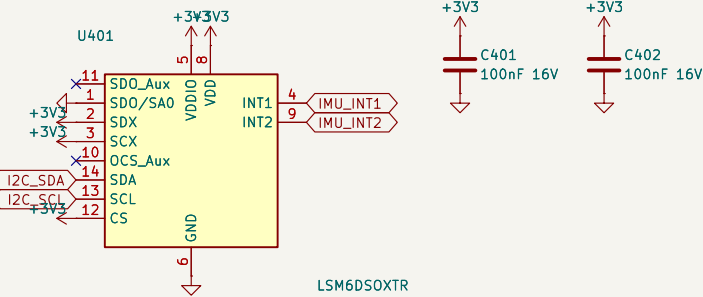
Rev: A0

KiCad E.D.A. 10.0.6

Id: 4/6

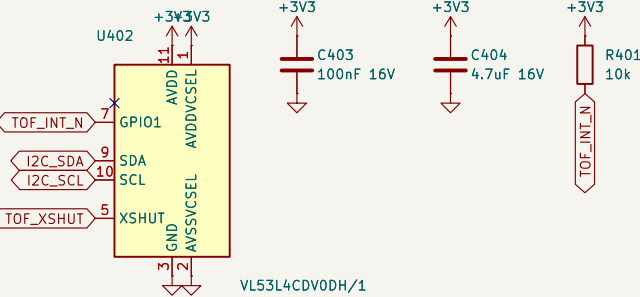
SENSE

IMU



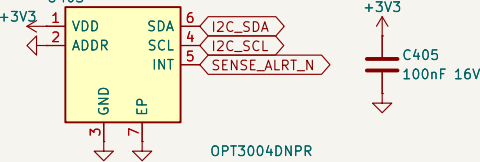
LSM6DSOX at 0x6A (SA0 low), I2C (CS high), SDx/SCx to VDDIO keeps the land compatible with the LSM6DS3TR-C fallback. INT1/INT2 push-pull, configured active-low for ext1 wake.

PROXIMITY



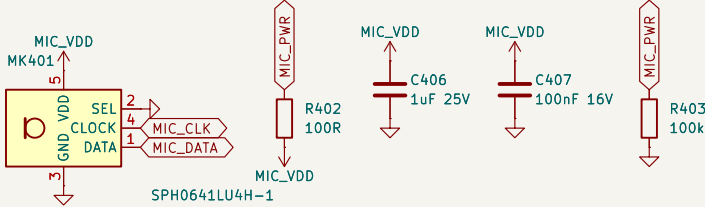
VL53L4CD time-of-flight, 0-1.3 m, under the window border. XSHUT from the expander (off by default); GPIO1 open-drain with the 10k pull-up ST recommends. AVDD decoupled 4.7 uF + 100 nF per ST.

AMBIENT LIGHT



OPT3004 at 0x44, human-eye response, under the window border. INT joins SENSE_ALERT_N.

MICROPHONE



SPH0641 bottom-port PDM mic on B.Cu, port through the board (0.5 mm hole) to the top. Powered from GPIO35 through 100R/1 uF only while listening: it draws 80 uA even with the clock stopped. SEL low.

RAIL SOURCE DECLARATIONS

IMU, proximity, ambient light, microphone

ODD JOBS

Sheet: /Sense/
File: sense.kicad_sch

Title: MAO_MAIN A0 - Sense

Size: A3
KiCad E.D.A. 10.0.6

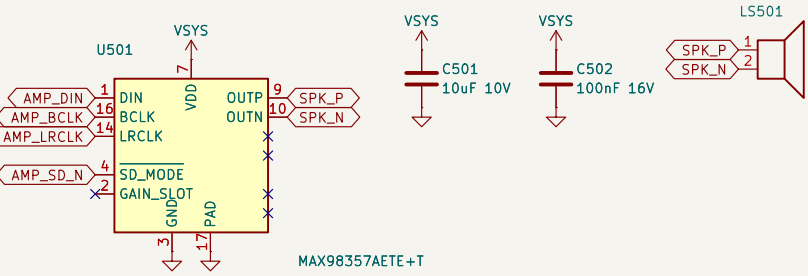
Date: 2026-01-06

Rev: A0

Id: 5/6

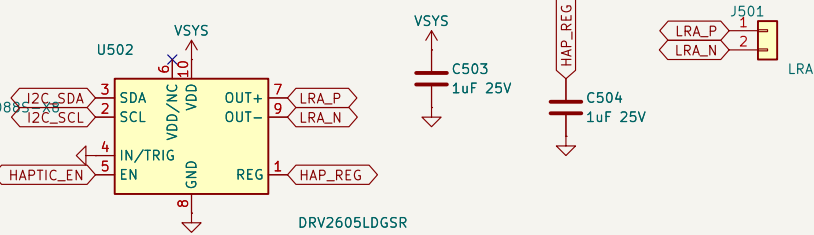
FEEDBACK

SPEAKER AMP



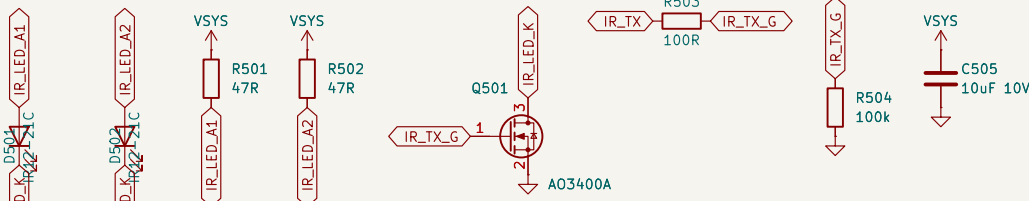
MAX98357A 12S class-D from VSYS. SD_MODE from the expander; low = 0.6 uA shutdown. 3.3 V = left channel. GAIN_SLOT open = 9 dB (datasheet state, deliberate). Speaker: Same Sky CMS-150803-088s-X8 (15 x 8 x 3 mm, 8 ohm, 0.8 W) in the base at 9 o'clock; its own spring contacts press on two pads on B.Cu (LS501).

HAPTIC



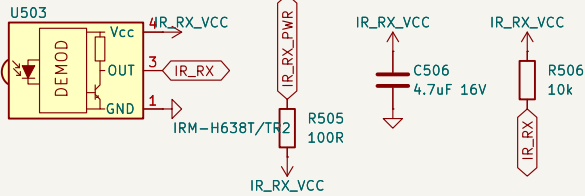
DRV2605L closed-loop LRA driver from VSYS. I2C 0x5A. EN from the expander (4 uA off). IN/TRIG low; playback by I2C. Pin 6 VDD/NC left open (SLOS854D; optional supply, tie to VDD or float). LRA LD0832AA (235 Hz, 1.8 Vrms) glued to the base, leads to J501.

IR TRANSMIT



Two side-emitting 940 nm LEDs at the back edge (B.Cu), each with its own 47R from VSYS (42-66 mA pulses at 38 kHz, 33 % duty, within the 65 mA rating). AO3400A low side, gate pulled down: dark at reset (117).

IR RECEIVE



IRM-H638T (as LCDkit), top-view, looking up through the window border. Powered from expander P5 through 100R/4.7 uF only when listening (0.4 mA). OUT pull-up goes to the switched supply, never to +3V3, so an unpowered receiver is never back-powered and GPIO40 reads a defined low while it is off.

RAIL SOURCE DECLARATIONS

Speaker amplifier, haptic driver, IR transmit and receive

ODD JOBS

Sheet: /Feedback/
File: feedback.kicad_sch

Title: MAO_MAIN A0 - Feedback

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-01-06
Pwr File: IR_RX_VCC

Rev: A0
Id: 6/6